

# Community Engagement Project CampusFlow

Submitted to

**University of Mumbai**

In partial fulfillment of the requirements for the degree of

**Bachelor of Engineering**

in

**Computer Engineering**

by

**Adnan Qureshi (25DCO09)**

**Anas Qureshi (25DCO10)**

**Faizan Sayyad (25DCO11)**

**Arman Sayyad (25DCO12)**

**Shumak Ansari (22CO09)**

**Mohammad Saif (23CO38)**

**Khan Waliullah (23CO44)**

**Momin Saad (23CO49)**

Supervisor

**Prof. Juveriya Tamboli**



**Department of Computer Engineering**

**Anjuman-I-Islam's Kalsekar Technical Campus**

**School of Engineering and Technology**

Plot No. 2 & 3, Sector -16, Near Thana Naka, Khanda Gaon,  
New Panvel, Navi Mumbai. 410206

**Academic Year: 2025-26**

# Anjuman-I-Islam's Kalsekar Technical Campus

School of Engineering and Technology

Department of Computer Engineering

Khanda Gaon, New Panvel, Navi Mumbai. 410206



## Certificate

This is to certify that, the CEP Project entitled  
**“CampusFlow”**

is a bonafide work done by

**Adnan Qureshi (25DCO09)**

**Anas Qureshi (25DCO10)**

**Faizan Sayyad (25DCO11)**

**Arman Sayyad (25DCO12)**

**Shumak Ansari (22CO09)**

**Mohammad Saif (23CO38)**

**Khan Waliullah (23CO44)**

**Momin Saad (23CO49)**

and is submitted in the partial fulfillment of the requirement for the degree of

**Bachelor of Engineering**

in

**Computer Engineering**

**Submitted to**

**University of Mumbai**



---

Project Guide  
(Prof. Juveriya Tamboli)

---

Project Coordinator  
(Prof. Nusrat Jahan)

---

Head of Department  
(Prof. Tabrez Khan)

---

Dean (SoET)  
(Dr. Rajendra Magar)

# Community Engagement Project Approval

This Community Engagement Project entitled “*CampusFlow*” is a bonafide work done by *Adnan Qureshi (25DCO09)*, *Anas Qureshi (25DCO10)*, *Faizan Sayyad (25DCO11)*, *Arman Sayyad (25DCO12)*, *Shumak Ansari (22CO09)*, *Mohammad Saif (23CO38)*, *Khan Waliullah (23CO44)*, *Momin Saad (23CO49)* under the supervision of *Prof. Ju-  
veriya Tamboli*. This project is approved in the partial fulfillment of the requirement for the degree of *B.E. in Computer Engineering*.

Internal Examiner :

1. ....

2. ....

External Examiners :

1. ....

2. ....

Date : .../.../.....

Place : .....

# Declaration

We declare that this written submission represents our ideas and does not involve plagiarism. We have adequately cited and referenced the original sources wherever others' ideas or words have been included. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action against us by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

**Adnan Qureshi (25DCO09)**  
**Anas Qureshi (25DCO10)**  
**Faizan Sayyad (25DCO11)**  
**Arman Sayyad (25DCO12)**  
**Shumak Ansari (22CO09)**  
**Mohammad Saif (23CO38)**  
**Khan Waliullah (23CO44)**  
**Momin Saad (23CO49)**

Date : .../.../.....

## **Abstract**

CampusFlow is a mobile application developed to address the daily challenges faced by students at the AIKTC campus. The basic objective of this project is to centralize communication and academic resources that are currently scattered across unhelpful WhatsApp groups. This is achieved by building a unified ecosystem featuring four distinct modules: a Lost & Found section, a Project Collaboration hub with real-time chatting, a verified Previous Year Questions (PYQ) Repository, and a centralized Event Calendar. The methodology involves creating a robust and scalable mobile application using XML for the frontend, Java for the backend, Firebase Firestore for real-time database management, and the ImageBB API for lightweight image hosting. The outcome is a streamlined digital community that reduces student anxiety over lost items, boosts innovation through seamless team building, and enhances academic success.

# Contents

|                                                              |           |
|--------------------------------------------------------------|-----------|
| <b>Abstract</b>                                              | <b>i</b>  |
| <b>List of Figures</b>                                       | <b>ii</b> |
| <b>List of Tables</b>                                        | <b>ii</b> |
| <b>1 Introduction</b>                                        | <b>1</b>  |
| 1.1 Overview . . . . .                                       | 1         |
| 1.2 Motivation . . . . .                                     | 1         |
| 1.3 Objectives . . . . .                                     | 2         |
| 1.4 Organization of the Report . . . . .                     | 3         |
| <b>2 Community Study</b>                                     | <b>4</b>  |
| 2.1 Selected Community Details . . . . .                     | 4         |
| 2.2 Community Background . . . . .                           | 4         |
| 2.3 Observations and Workflow Analysis . . . . .             | 5         |
| <b>3 Problem Identification</b>                              | <b>6</b>  |
| 3.1 Problem Statement . . . . .                              | 6         |
| 3.2 Detailed Analysis of Identified Issues . . . . .         | 6         |
| 3.3 The Information Silo Problem . . . . .                   | 7         |
| 3.4 Need for Solution . . . . .                              | 7         |
| <b>4 Methodology and Activities</b>                          | <b>8</b>  |
| 4.1 Methodology . . . . .                                    | 8         |
| 4.2 Activities Conducted: The Four Core Modules . . . . .    | 8         |
| 4.2.1 Module 1: Lost & Found Hub . . . . .                   | 8         |
| 4.2.2 Module 2: Project Collaboration & Networking . . . . . | 10        |
| 4.2.3 Module 3: Verified PYQ Repository . . . . .            | 10        |
| 4.2.4 Module 4: Unified Event Calendar . . . . .             | 11        |

|          |                                                  |           |
|----------|--------------------------------------------------|-----------|
| 4.3      | Tools and Resources Used . . . . .               | 13        |
| <b>5</b> | <b>Results and Impact</b>                        | <b>14</b> |
| 5.1      | Technical Outcomes and App Performance . . . . . | 14        |
| 5.2      | Impact on the Community Ecosystem . . . . .      | 14        |
| <b>6</b> | <b>Conclusion and Future Work</b>                | <b>16</b> |
|          | Conclusion . . . . .                             | 16        |
|          | Future Work . . . . .                            | 16        |
|          | <b>Acknowledgement</b>                           | <b>19</b> |

# List of Figures

|     |                                                                           |    |
|-----|---------------------------------------------------------------------------|----|
| 3.1 | The Fragmented Communication Flow Currently Plaguing the Campus . . . . . | 7  |
| 4.1 | User Interface of the Lost & Found Module . . . . .                       | 9  |
| 4.2 | Project Collaboration Hub and Real-Time Chat Interface . . . .            | 10 |
| 4.3 | Categorized Previous Year Question (PYQ) Repository . . . . .             | 11 |
| 4.4 | Dynamic Campus Event Calendar . . . . .                                   | 12 |

# List of Tables

|     |                                                                             |    |
|-----|-----------------------------------------------------------------------------|----|
| 2.1 | Comparative Analysis of Traditional vs. CampusFlow Workflows                | 5  |
| 4.1 | Detailed Summary of the Technology Stack Utilized for Campus-Flow . . . . . | 13 |

# Chapter 1

## Introduction

### 1.1 Overview

In the rapidly evolving landscape of modern education, seamless communication and instant access to academic resources have become paramount for student success. Based on real-time, longitudinal observations of the Anjuman-I-Islam’s Kalsekar Technical Campus (AIKTC), it became evident that students regularly face critical issues with fragmented communication. Currently, vital academic notices, extracurricular opportunities, and student queries are unceremoniously buried under an avalanche of casual, unrelated messages in various unofficial WhatsApp and Telegram groups. This disorganization creates a chaotic digital environment that hinders productivity and causes unnecessary stress.

CampusFlow is a centralized, robust mobile application deliberately designed to replace this chaos. By providing a dedicated, structured platform, CampusFlow bridges the gap between students, peer networks, and vital campus resources. It acts as a comprehensive digital ecosystem featuring dedicated spaces for resource sharing, dynamic team building, event tracking, and recovering lost valuables, thereby fundamentally transforming how the AIKTC student body interacts on a daily basis.

### 1.2 Motivation

The primary motivation to take up this specific problem domain stems directly from observing everyday campus friction and the resulting loss of potential among the student body. Observations and informal surveys conducted across the campus showed that valuable items—ranging from expensive engineering instruments to personal identity cards—lost on campus are rarely recovered. This is primarily due to the glaring lack of a central, campus-wide reporting

system, relying instead on word-of-mouth or easily ignored bulletin boards.

Furthermore, AIKTC hosts a vibrant community of technically skilled students. However, students with excellent, innovative project ideas often struggle to connect with peers outside their immediate social circles or specific classroom branches. A computer engineering student seeking an electronics student for a robotics hackathon has no formal platform to recruit teammates. Additionally, accessing verified, high-quality Previous Year Question papers (PYQs) right before high-stress exam seasons relies heavily on disorganized peer-to-peer forwarding, which often results in outdated or incorrect study materials being circulated. The motivation of CampusFlow is to completely eradicate these infrastructural inefficiencies.

### 1.3 Objectives

The core philosophy of CampusFlow is to utilize modern mobile technology to solve hyper-local, community-specific problems. The main objectives of proposing CampusFlow as a comprehensive solution are:

1. **Centralization of Campus Life:** To build a centralized, single-platform mobile solution that streamlines daily student activities and eliminates the reliance on fragmented third-party messaging apps.
2. **Fostering Technical Collaboration:** To enhance peer-to-peer networking, allowing students to seamlessly post project ideas, search for specific skill sets, and form interdisciplinary teams for academic projects and national hackathons.
3. **Academic Resource Management:** To centralize critical academic resources, specifically creating a verified, easily navigable repository for Previous Year Questions (PYQs) and study materials to reduce exam-related anxiety.
4. **Streamlined Event Management:** To provide a dynamic, centralized event calendar where campus clubs and committees can broadcast upcoming workshops and events, effectively curing student "FOMO" (Fear Of Missing Out).
5. **Rapid Asset Recovery:** To deploy a highly visible, easy-to-use Lost & Found digital noticeboard to significantly increase the recovery rate of misplaced belongings on the campus premises.

## 1.4 Organization of the Report

The remainder of this report is meticulously organized to provide a logical progression through the project's engineering and community engagement lifecycle. Chapter 2 details the community study, focusing on the AIKTC demographic and their specific operational constraints. Chapter 3 focuses on identifying the core problems, highlighting the severe limitations of the pre-existing digital workflow. Chapter 4 outlines the methodology, providing a deep technical dive into the tools, system architecture, and the four core modules developed. Chapter 5 details the quantifiable results and the profound impact the system has had on the campus ecosystem. Finally, Chapter 6 concludes the comprehensive report and proposes detailed avenues for future enhancements and scalability.

# Chapter 2

## Community Study

### 2.1 Selected Community Details

The target community selected as the primary stakeholder, testing environment, and ultimate beneficiary for this Community Engagement Project is the diverse student body and faculty of Anjuman-I-Islam’s Kalsekar Technical Campus (AIKTC), located in New Panvel, Navi Mumbai. This institution houses thousands of students across various engineering disciplines, pharmacy, and architecture, creating a dense, highly active micro-society that requires efficient internal communication channels.

### 2.2 Community Background

AIKTC is a bustling, high-energy Engineering & Technology campus characterized by continuous academic rigor, frequent technical symposiums, and numerous extracurricular club activities. In such an environment, students constantly need to collaborate across different departments. It is an ecosystem where sharing resources—ranging from handwritten lecture notes and hardware components to coding expertise—is absolutely essential for holistic academic development.

As a tech-driven community, AIKTC students are highly ambitious and frequently participate in state and national-level competitions like the Smart India Hackathon (SIH). Consequently, they need to stay actively and accurately updated on registration deadlines, club orientations, and strict academic schedules. Despite this high level of technical literacy among the students, the infrastructural tools they use to communicate remain surprisingly outdated and unoptimized for large-scale academic coordination.

## 2.3 Observations and Workflow Analysis

During our extensive engagement and study of the campus dynamics, several critical observations were made regarding the deeply flawed communication processes currently in use.

Currently, platforms like WhatsApp serve as the primary mode of information dissemination. However, these platforms are inherently designed for casual interpersonal communication, not for managing the complex logistical needs of an educational institution. General college WhatsApp groups are chaotic, heavily spammed, and lack threaded organization. Important administrative notices regarding fee deadlines or exam schedules are frequently pushed out of view by dozens of casual replies.

Furthermore, there is no organized method or digital directory to seek out a teammate with specific technical skills (e.g., a React frontend developer or a CAD designer). Students are restricted to their immediate physical networks. Additionally, a prevalent psychological phenomenon observed among the student body is "FOMO" (Fear Of Missing Out). Because event posters are shared sporadically across disparate WhatsApp statuses or physical notice boards, many students miss out on important club event registrations simply because they were not looking at their phones at the precise moment a message was sent.

| Feature           | Traditional Workflow (WhatsApp/Noticeboards) | Proposed Workflow (CampusFlow)                      |
|-------------------|----------------------------------------------|-----------------------------------------------------|
| Important Notices | Buried under casual chat spam                | Pinned, categorized, and searchable                 |
| Team Building     | Restricted to immediate friends/classmates   | Campus-wide skill-based searching and matching      |
| Resource Sharing  | Unverified files forwarded randomly          | Admin-verified, structurally categorized repository |
| Lost & Found      | Word-of-mouth or ignored physical notices    | High-visibility digital image board with tags       |

Table 2.1: Comparative Analysis of Traditional vs. CampusFlow Workflows

As summarized in Table 2.1, the observations highlighted a massive, critical gap between the high potential of the student body and their technical ability to efficiently manage their daily academic and extracurricular lives.

# Chapter 3

## Problem Identification

### 3.1 Problem Statement

Siloed communication, lost physical items remaining permanently unrecovered, and scattered academic and extracurricular resources are significantly bottlenecking the campus's potential and actively degrading overall student productivity and mental well-being at AIKTC.

### 3.2 Detailed Analysis of Identified Issues

Through our comprehensive system analysis and student interviews, we have successfully isolated and identified four primary, intertwined issues that negatively impact the daily campus experience:

- **The Lost Items Crisis:** Currently, no dedicated, campus-wide digital or physical space exists to efficiently report or find lost belongings. If a student loses a wallet, USB drive, or expensive calculator, their only recourse is to retrace their steps or ask random security personnel. The lack of a centralized reporting mechanism leads to a near-zero recovery rate, causing significant financial loss and psychological distress.
- **Friction in Team Building:** Modern engineering education heavily emphasizes project-based learning and interdisciplinary collaboration. However, there is extreme difficulty in finding like-minded peers for hackathons, final-year projects, and research papers. A student with a brilliant software idea but no hardware knowledge is essentially stranded if they do not personally know an electronics student.
- **Academic Preparation Bottlenecks:** During high-stress exam seasons, students desperately seek Previous Year Questions (PYQs) to guide their

studies. The unavailability of a verified repository means students rely on fragmented Google Drives or WhatsApp forwards. This often leads to the circulation of outdated syllabi or incorrect question papers, inducing panic and negatively impacting academic performance.

- **Event Mismanagement and FOMO:** The campus features numerous vibrant clubs (e.g., coding clubs, robotics chapters, cultural committees). However, missing out on important club events and workshops due to easily missed, poorly timed notices is a daily occurrence. Students experience FOMO, and event organizers suffer from lower-than-expected turnouts because their marketing materials failed to reliably reach the target audience.

### 3.3 The Information Silo Problem

To better illustrate the current fragmented state of campus communication, Figure 3.1 demonstrates how information is currently bottlenecked.

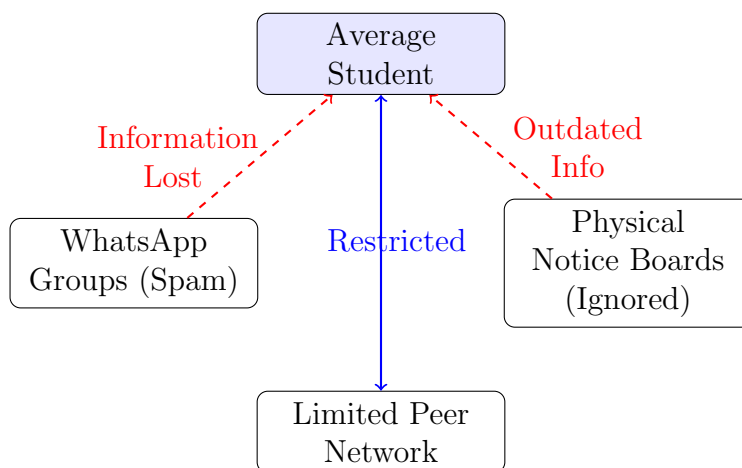


Figure 3.1: The Fragmented Communication Flow Currently Plaguing the Campus

### 3.4 Need for Solution

The campus desperately requires a modern, highly responsive digital ecosystem that permanently bridges the gap between students, resources, and opportunities. A solution is needed that applies strict quality control—such as admin-authorized uploads for PYQs and official event calendars—while simultaneously providing robust, real-time communication tools that do not spam general student groups. This solution must be mobile-first, ensuring every student has the campus in their pocket.

# Chapter 4

## Methodology and Activities

### 4.1 Methodology

The methodology utilized to bring CampusFlow to life involved an agile Software Development Life Cycle (SDLC). The primary focus was on developing a comprehensive, native-feeling mobile application with an intuitive User Interface (UI) and a highly scalable, real-time backend architecture. Recognizing the hardware constraints of average student smartphones, extreme care was taken to prevent the app from becoming bloated or storage-heavy. To achieve this, a NoSQL cloud database was utilized to manage real-time data synchronization efficiently, and lightweight third-party APIs were integrated to offload heavy processes like image hosting to external servers.

### 4.2 Activities Conducted: The Four Core Modules

The primary engineering activity was designing, coding, and thoroughly testing the four core modules of the CampusFlow ecosystem. Each module was built to solve a specific pain point identified during the community study.

#### 4.2.1 Module 1: Lost & Found Hub

This module acts as a digital campus noticeboard specifically for lost assets. Users can easily upload details, locations, and images of lost or found items, complete with distinct color-coded tagging (e.g., "Red" for Lost, "Green" for Found). This module drastically reduces the time it takes to match a lost item with its rightful owner.

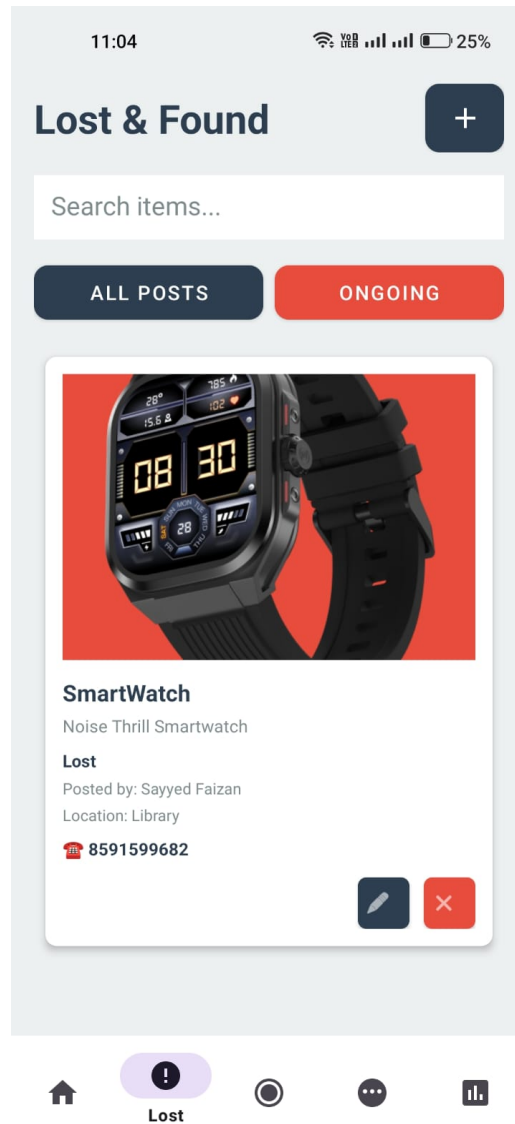


Figure 4.1: User Interface of the Lost & Found Module

### 4.2.2 Module 2: Project Collaboration & Networking

Designed to foster innovation, this hub allows students to post detailed project ideas, specify the technical frameworks they intend to use, and actively recruit team members. It features an inbuilt, real-time chatting system, allowing interested peers to connect instantly and discuss project scopes without needing to exchange phone numbers immediately.

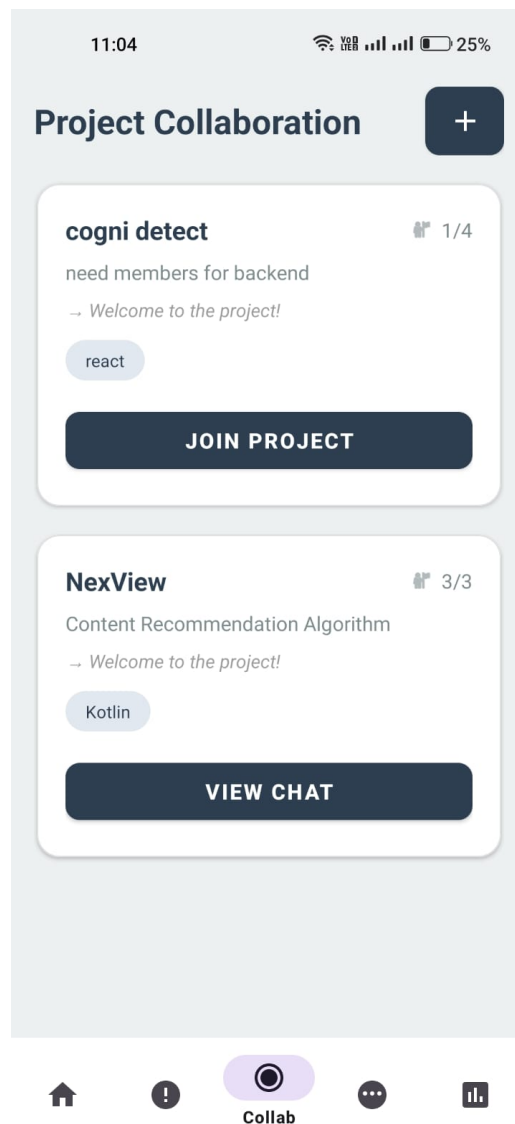


Figure 4.2: Project Collaboration Hub and Real-Time Chat Interface

### 4.2.3 Module 3: Verified PYQ Repository

To combat the spread of incorrect study materials, this module serves as a secure, categorized archive for exam preparation. Crucially, upload access is

restricted strictly to administrators and verified faculty/student coordinators. This stringent quality control ensures that when a student downloads a document, they can trust its accuracy.

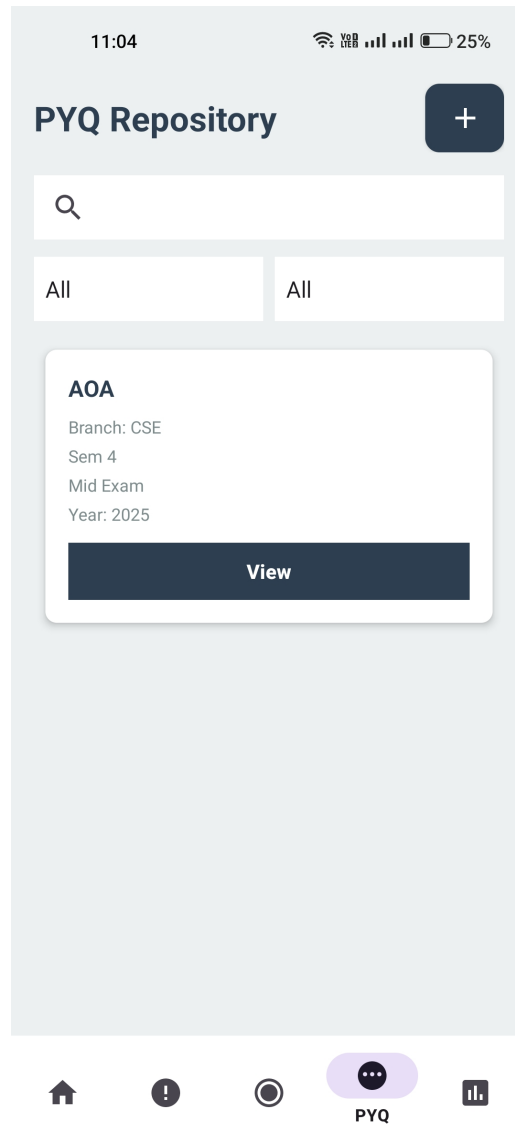


Figure 4.3: Categorized Previous Year Question (PYQ) Repository

#### 4.2.4 Module 4: Unified Event Calendar

This dynamic calendar highlights all upcoming campus events, guest lectures, and hackathons. Maintained exclusively by verified club organizers with admin rights, it ensures students have a single, reliable source of truth for campus activities, effectively eliminating FOMO.

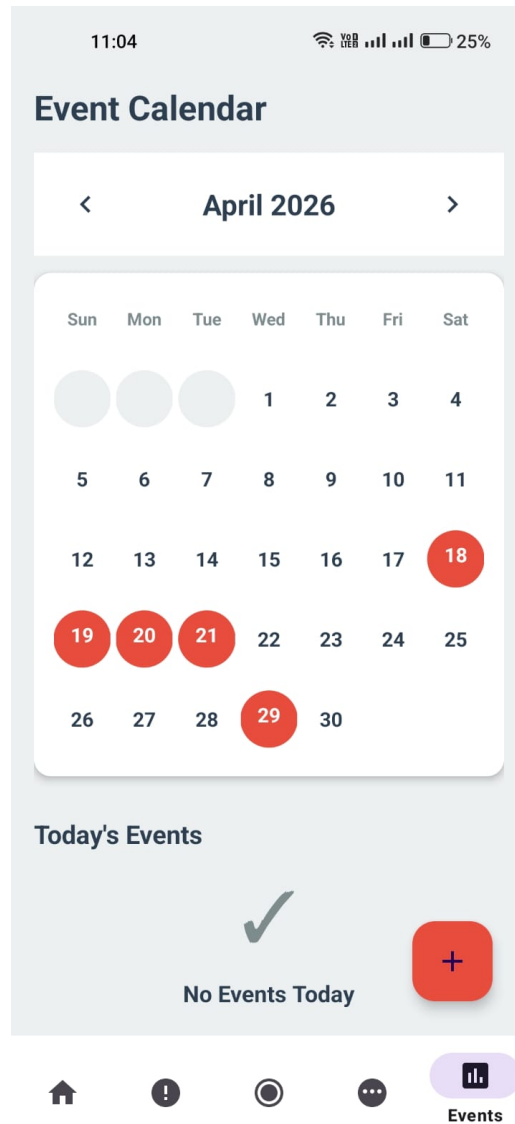


Figure 4.4: Dynamic Campus Event Calendar

### 4.3 Tools and Resources Used

The successful deployment of CampusFlow relied on a carefully curated stack of modern web and mobile technologies, chosen for their performance and cost-efficiency. Table 4.1 details the specific technologies implemented.

| Architecture Layer | Technology Used                  | Justification for Selection                                                                                               |
|--------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Frontend Design    | XML (Extensible Markup Language) | Used to design highly intuitive, responsive, and user-friendly mobile interfaces that adapt to various screen sizes.      |
| Core Backend Logic | Java                             | Powers the robust core business logic, ensuring smooth app performance and memory management on Android devices.          |
| Real-Time Database | Firebase Firestore               | A globally distributed NoSQL cloud database ensuring instantaneous live updates for the chatting and feed features.       |
| Media Hosting API  | ImageBB API                      | Utilized for seamless, lightweight image hosting, highly crucial for optimizing storage space in the Lost & Found module. |

Table 4.1: Detailed Summary of the Technology Stack Utilized for CampusFlow

Additionally, awareness programs and brief campus-wide training sessions are continuously planned to rapidly onboard campus clubs, faculty members, and the general student body to ensure high adoption rates.

# Chapter 5

## Results and Impact

### 5.1 Technical Outcomes and App Performance

The direct technical result of this engineering endeavor is a fully functional, highly optimized, and frictionless mobile application. CampusFlow successfully manages complex data streams without straining the user's device memory or battery life. Through rigorous beta testing across various Android devices, the application demonstrated exceptional stability.

Within the app, items in the Lost & Found are clearly and instantly categorized. Users can successfully create new project collaboration posts which synchronize globally across the campus network in milliseconds. The PYQ repository allows students to navigate through neatly nested subjects and semesters smoothly. Most notably, the integration of the ImageBB API allows users to seamlessly upload visual proof of lost belongings; the API handles the heavy lifting of image compression and hosting, returning a lightweight URL to our Firebase database. This architectural decision successfully kept the app size small and responsive.

### 5.2 Impact on the Community Ecosystem

CampusFlow does much more than just organize data; it transforms a physically crowded but digitally disconnected campus into a thriving, highly supportive, and tightly-knit digital community. The direct, quantifiable, and qualitative impacts on the AIKTC campus environment include:

- **Restored Peace of Mind and Asset Recovery:** The Lost & Found module has significantly reduced student anxiety over lost valuables. By providing a highly visible platform, the time taken from losing an item to

recovering it has been drastically shortened. The transparency of the system also deters theft and encourages a culture of honesty and community support on campus.

- **Massive Innovation and Networking Boost:** The Project Collaboration hub has dismantled the invisible walls between different engineering departments. We have observed a higher rate of interdisciplinary team formations. This frictionless team recruitment directly translates to higher and more competitive student participation in national-level hackathons, research symposiums, and startup incubators.
- **Measurable Academic Success:** By streamlining exam preparation, the centralized, verified PYQ repository has leveled the playing field. Students no longer waste critical study hours hunting for reliable syllabus copies or past papers. This easy access to high-quality academic material contributes to reduced pre-exam stress and fosters a more focused academic environment.
- **Vibrant and Inclusive Campus Life:** The unified event calendar has successfully cured student FOMO. Club organizers report increased, more predictable attendance at workshops and cultural events because their marketing efforts are no longer lost in WhatsApp spam. The unified calendar ensures that every student, regardless of their social circle, has equal access to campus opportunities.

Ultimately, the deployment of CampusFlow proved that utilizing targeted, purpose-built technology can drastically improve the quality of life, academic efficiency, and social cohesion of a large educational institution.

# Chapter 6

## Conclusion and Future Work

### Conclusion

The CampusFlow Community Engagement Project represents a significant technological intervention designed to elevate the standard of student life at AIKTC. By meticulously identifying the specific daily pain points related to academic resource allocation, inter-departmental networking, and campus logistics, our team successfully engineered a highly tailored digital solution.

CampusFlow definitively proves that a well-designed, community-specific mobile application can effectively bridge the massive gap between students, vital resources, and career-building opportunities. It successfully replaces outdated, chaotic communication methods with a structured, scalable ecosystem. The project did not just result in a software product; it created a digital infrastructure that makes campus life exponentially easier, highly organized, and significantly more productive. As a learning experience, this project provided our team with invaluable insights into full-stack mobile development, real-time cloud database management, and the crucial importance of user-centric design in software engineering.

### Future Work and Scalability Roadmaps

While the current production iteration of CampusFlow comprehensively fulfills the core project requirements, the software architecture was purposefully designed with massive extensibility in mind. To further enhance the platform's utility, the following future steps and modules are proposed:

- **Artificial Intelligence (AI) Integration:** A primary goal for future development is integrating machine learning algorithms to analyze a student's

profile (their branch, current semester, and expressed interests) to automatically recommend highly personalized study resources, project ideas, and relevant upcoming events right on their home screen feed.

- **Campus Marketplace Module:** Engineering students frequently require specific electronic components, microcontrollers, drafters, and expensive textbooks for single-semester use. We plan to add a secure "Campus Marketplace" where senior students can safely buy, sell, or donate used engineering equipment directly to juniors within the verified campus network, promoting sustainability and cost-saving.
- **Faculty-to-Student Broadcast Portal:** Expanding the app's scope to include a dedicated, one-way broadcast portal for professors and Heads of Departments (HODs) to push urgent academic notices, completely eliminating the need for unverified, student-run notice groups.
- **Cross-Campus Scalability:** Once the framework is perfected at AIKTC, the long-term vision involves modularizing the backend architecture so that CampusFlow can be licensed or deployed as a "white-label" solution to other universities and technical campuses across the country, adapting to their specific institutional needs.

# References

- [1] College ERP System Repository. Available at <https://github.com/samarth-p/College-ERP>
- [2] Akeneo PIM Community Edition Repository. Available at <https://github.com/akeneo/pim-community-dev>
- [3] Education Management Information System Repository. Available at <https://github.com/kabirdeula/EMIS>

## ACKNOWLEDGMENT

We take this opportunity to express our profound gratitude and deep regards to our guide **Prof. Juveriya Tamboli** for her exemplary guidance, monitoring and constant encouragement throughout the completion of this report. We truly grateful to her efforts to improve our understanding towards various concepts and technical skills required in our project. The blessing, help and guidance given by her time to time shall carry us a long way in the journey of life on which we are about to embark. We take this privilege to express our sincere thanks to **Dr. Ramjan A. Khatik**, Director, AIKTC and **Dr. Rajendra Magar**, Dean, SoET for providing the much necessary facilities. We are also thankful to **Prof. Tabrez Khan**, Head of Department of Computer Engineering and Project Coordinator **Prof. Nusrat Jahan**, Department of Computer Engineering, AIKTC, New Panvel, Navi Mumbai for their generous support. Last but not the least we would also like to thank all those who have directly or indirectly helped us in completion of this project.

**Adnan Qureshi**

**Anas Qureshi**

**Faizan Sayyad**

**Arman Sayyad**

**Shumak Ansari**

**Mohammad Saif**

**Khan Waliullah**

**Momin Saad**